Reliable Machine Learning with Unreliable Black Boxes **Spring 2026**

Lecture 4: Conformal p-values, Coverage, and Distributional CP

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1 Recap: Split Conformal Prediction

We begin with a brief recap of the *split conformal prediction* (split CP) procedure, which provides distribution-free predictive coverage guarantees given access to a black-box predictor $\hat{f}$.

Suppose we have access to a dataset which we partition into two parts:

- n **training samples** $(X_1, Y_1), \ldots, (X_n, Y_n)$, used only to fit the model;
- n **calibration samples** $(\bar{X}_1, \bar{Y}_1), \ldots, (\bar{X}_n, \bar{Y}_n)$, which are exchangeable with the test point (X_{n+1}, Y_{n+1}).

The split CP algorithm proceeds in three steps:

(1) Score function construction. Use the training samples $(X_1, Y_1), \ldots, (X_n, Y_n)$ to learn a *score function* $S(x, y)$. We require S to be *symmetric*, i.e., invariant under permutations of the calibration points, so that exchangeability transfers from the data to the scores. A canonical example is the absolute residual score,

$$S(x, y) = |y - \hat{f}(x)|,$$

where $\hat{f}$ is the model trained on the training split.

(2) Calibration. Compute the scores on the calibration samples,

$$S_i := S(\bar{X}_i, \bar{Y}_i), \qquad i = 1, \ldots, n,$$

and form the empirical quantile

$$\hat{q}_{1-\alpha} \;=\; \lfloor (1-\alpha)(n+1) \rfloor\text{-th smallest of } S_1, S_2, \ldots, S_n.$$

(3) Prediction set. At a test point X_{n+1}, output

$$C(X_{n+1}) \;=\; \big\{\, y \in \mathcal{Y} \;:\; S(X_{n+1}, y) \le \hat{q}_{1-\alpha} \,\big\}.$$

Theorem 1 (Guaranteed marginal coverage)**.** *If the calibration and test samples*

$$(\bar{X}_1, \bar{Y}_1), \ldots, (\bar{X}_n, \bar{Y}_n),\ (X_{n+1}, Y_{n+1})$$

are exchangeable, then

$$\Pr\big[\, Y_{n+1} \in C(X_{n+1}) \,\big] \;\ge\; 1 - \alpha.$$

This guarantee is *distribution-free*: it relies only on exchangeability of the calibration and test points, not on correctness of $\hat{f}$ or knowledge of the underlying distribution.

2 Hypothesis Testing for Exchangeability

Split CP is justified under an exchangeability assumption. A natural question is: can we *test* whether a given dataset is exchangeable? This turns out to be tightly linked to the machinery of conformal p-values, and uses the classical idea of *permutation/rank tests*.

2.1 Setup

Given a sample from some distribution $\mathcal{D}$,

$$(Z_1, Z_2, \ldots, Z_{n+1}) \sim \mathcal{D},$$

we consider the hypothesis test

$$H_0 : \mathcal{D} \in \mathcal{P}_{\text{exchangeable}} \quad \text{vs.} \quad H_1 : \mathcal{D} \in \mathcal{P}_{\text{non-exchangeable}},$$

where $\mathcal{P}_{\text{exchangeable}}$ denotes the class of all exchangeable distributions over $(n+1)$-tuples.

Remark 2. A test for this hypothesis cannot in general have high power against *every* non-exchangeable alternative. There exist non-exchangeable distributions that are nearly indistinguishable from exchangeable ones using only a single sample, so such $\mathcal{D}$ could be non-exchangeable but still pass the test. This limitation is inherent rather than a shortcoming of a particular procedure.

2.2 Desiderata for a Test

We want a test statistic $p(Z_1, Z_2, \ldots, Z_{n+1}) \in [0, 1]$ satisfying:

(1) p is *low* ideally when Z_{n+1} is an "outlier" relative to $Z_1, \ldots, Z_n$ (high power against natural alternatives).

(2) p is *high* when $(Z_1, \ldots, Z_{n+1}) \sim \mathcal{D} \in \mathcal{P}_{\text{exch}}$. More precisely, under exchangeability,

$$\Pr\big[\, p(Z_1, \ldots, Z_{n+1}) \le \alpha \,\big] \le \alpha \qquad \text{(super-uniform).}$$

2.3 The Conformal p-value

A simple statistic meeting these desiderata is the (rank-based) *conformal p-value*:

$$p(Z_1, \ldots, Z_{n+1}) \;=\; \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\big\{\, Z_i \ge Z_{n+1} \,\big\}.$$

This is simply the (suitably normalized) rank of Z_{n+1} among all $n+1$ values, counting from the top. If Z_{n+1} is unusually large (an outlier on the right), the indicator is only satisfied at $i = n+1$, pushing p toward $\frac{1}{n+1}$.

Theorem 3 (Validity of the conformal p-value)**.** *The statistic p defined above is a valid p-value for exchangeability:*

$$\Pr_{(Z_1,\ldots,Z_{n+1}) \sim \textit{Exchangeable}} \big(p(Z_1, \ldots, Z_{n+1}) \le \alpha\big) \;\le\; \alpha \qquad \textit{(super-uniformity).}$$

3 Conformal p-values

To prove Theorem 3, we use two elementary facts about CDFs.

Fact 4 (Continuous case)**.** *Let X be any (absolutely) continuous real-valued random variable with CDF F_X, and let $T \sim X$. Then*

$$F_X(T) \sim \text{Uniform}[0,1].$$

Fact 5 (General case)**.** *Let X be* any *real-valued random variable with CDF F_X, and let $T \sim X$. Then $F_X(T)$ is* super-uniform*: for any random variable q having the same distribution as $F_X(T)$,*

$$\Pr\big[\, q \le \alpha \,\big] \;\le\; \alpha, \qquad \forall \alpha \in [0,1].$$

Fact 5 is the one we actually need, because the empirical distribution we plug in below is discrete.

Proof of Theorem 3. Conditional on the unordered collection of values $\{Z_1, \ldots, Z_{n+1}\}$, let $\mathcal{Q}$ be a discrete random variable that takes each of these values with equal probability:

$$\mathcal{Q} \;=\; \begin{cases} Z_1 & \text{w.p. } \frac{1}{n+1}, \\ Z_2 & \text{w.p. } \frac{1}{n+1}, \\ \vdots & \\ Z_{n+1} & \text{w.p. } \frac{1}{n+1}. \end{cases}$$

Unpacking the definition of p, we see that

$$\begin{aligned} p(Z_1, \ldots, Z_{n+1}) \;&\equiv\; \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{Z_i \ge Z_{n+1}\} \\ &=\; \sum_{i=1}^{n+1} \Pr_{\mathcal{Q}}(\mathcal{Q} = Z_i) \cdot \mathbf{1}\{Z_i \ge Z_{n+1}\} \\ &=\; \Pr_{\mathcal{Q}}\big[\, \mathcal{Q} \ge Z_{n+1} \,\big], \end{aligned}$$

which is exactly the upper-tail CDF of $\mathcal{Q}$ evaluated at Z_{n+1}. Under exchangeability, all coordinates of $(Z_1, \ldots, Z_{n+1})$ are interchangeable: once we fix the collection of values, Z_{n+1} is equally likely to be any one of them. Hence Z_{n+1} has the same (conditional) distribution as $\mathcal{Q}$.

Applying Fact 5 to $-\mathcal{Q}$ (whose lower-tail CDF equals the upper-tail CDF of $\mathcal{Q}$) then gives super-uniformity:

$$\Pr\big[\, p(Z_1, \ldots, Z_{n+1}) \le \alpha \,\big] \;\le\; \alpha. \quad \square$$

$\square$

3.1 Conformal Prediction Sets via p-values

We now return to the prediction-set problem. Take

$$Z_i \;=\; S_i \;:=\; S(X_i, Y_i) \quad (i = 1, \ldots, n), \qquad S_{n+1}^{y} \;:=\; S(X_{n+1}, y).$$

Since S is a fixed (symmetric) score function and the underlying data are exchangeable, the scores $(S_1, \ldots, S_n, S_{n+1}^{Y_{n+1}})$ are themselves exchangeable. Detecting whether S_{n+1}^{y} is an "outlier" relative to $S_1, \ldots, S_n$ is therefore equivalent to deciding whether y belongs in the prediction set.

Concretely, define the *conformal p-value at candidate* y by

$$p^{y} \;:=\; \frac{1}{n+1} \sum_{i=1}^{n+1} \mathbf{1}\{S_i \geq S_{n+1}^{y}\},$$

with the convention $S_{n+1} := S_{n+1}^{y}$ inside the sum, and the *conformal prediction set*

$$C(X_{n+1}) \;:=\; \{\, y \in \mathcal{Y} \;:\; p^{y} > \alpha \,\}.$$

Applying Theorem 3 to $Z_i := S_i$, the p-value $p^{Y_{n+1}}$ is super-uniform, so

$$\Pr\big[\, Y_{n+1} \notin C(X_{n+1}) \,\big] \;=\; \Pr\big[\, p^{Y_{n+1}} \leq \alpha \,\big] \;\leq\; \alpha.$$

That is, p^{y} is a valid p-value for the hypothesis "$Y_{n+1} = y$," and the resulting set $C(X_{n+1})$ achieves $1 - \alpha$ marginal coverage.

4 Coverage Notions

Given $(X_1, Y_1), \ldots, (X_n, Y_n)$ and a new feature X_{n+1} (whose label Y_{n+1} is not observed), we output a prediction set $C(X_{n+1})$. There are several distinct notions of "coverage" one might ask for.

4.1 Marginal Coverage

Definition 6 (Marginal coverage)**.** A predictor C achieves *marginal coverage* at level $1 - \alpha$ if

$$\Pr_{\{(X_i,Y_i)\}_{i\in[n]},\,(X_{n+1},Y_{n+1})}\big[\, Y_{n+1} \in C(X_{n+1}) \,\big] \;\geq\; 1 - \alpha.$$

This is the guarantee delivered by split CP. The probability is over the joint draw of all $n + 1$ samples; in particular, coverage is averaged over both the randomness in the calibration set and in the test point.

4.2 Conditional Coverage

Definition 7 (Conditional coverage)**.** C achieves *conditional coverage* at level $1 - \alpha$ if

$$\Pr\big[\, Y_{n+1} \in C(X_{n+1}) \,\big|\, X_{n+1} \,\big] \;\geq\; 1 - \alpha \qquad \text{(almost surely in } X_{n+1}\text{)}.$$

Conditional coverage is a much stronger requirement: the set must cover the label with probability $\geq 1 - \alpha$ for (almost) *every* test point, not just on average. It is achievable in simple settings, e.g. when $X_{n+1} \in \mathcal{X} = \{a_1, \ldots, a_m\}$ is finite).

4.3 Impossibility of Non-Trivial Conditional Coverage

In the general continuous setting, however, conditional coverage forces the prediction set to be *vacuous*.

Theorem 8 (Impossibility of non-trivial conditional coverage)**.** *Let* $(X_i, Y_i) \overset{iid}{\sim} P$*. Any method* C *that achieves conditional coverage for* all *distributions* P *whose marginal* P_X *is (absolutely) continuous (i.e. admits a density) must output*

$$C(X_{n+1}) \;=\; \mathcal{Y} \qquad \textit{almost surely.}$$

In other words, distribution-free conditional coverage over continuous X is essentially equivalent to trivial prediction — the prediction set must be uninformatively large (literally $\mathcal{Y}$ when $\mathcal{Y}$ is bounded, and of infinite expected size more generally).

4.4 Weaker Relaxations

Between marginal and conditional coverage sit several practically useful relaxations.

Group coverage. Given groups $G_1, \ldots, G_m \subseteq \mathcal{X}$,

$$\forall j \in [m], \qquad \Pr\big[\, Y_{n+1} \in C(X_{n+1}) \,\big|\, X_{n+1} \in G_j \,\big] \;\geq\; 1 - \alpha.$$

This is often used for fairness-style guarantees across demographic subgroups.

Label-conditional coverage.

$$\Pr\big[\, Y_{n+1} \in C(X_{n+1}) \,\big|\, Y_{n+1} = y \,\big] \;\geq\; 1 - \alpha \qquad \forall y \in \mathcal{Y}.$$

This ensures balanced coverage across classes, which is important in imbalanced classification (e.g., rare disease detection).

Conditional coverage for "most" x. Relax conditional coverage to hold for a large fraction (e.g. $1 - \delta$) of test points X_{n+1}, rather than for all of them almost surely. This is the relaxation exploited by distributional conformal prediction, discussed next.

5 Distributional Conformal Prediction (DCP)

Distributional conformal prediction (DCP; Chernozhukov, Wüthrich, and Zhu [2021]) builds on conformalized quantile regression (CQR; Romano, Patterson, and Candès [2019]). The motivation is to move beyond predicting a point estimate $\hat{f}(x)$ and instead leverage an estimate of the full conditional distribution $Y \mid X$.

5.1 From Point Predictors to Distribution Predictors

Split CP, as stated, requires only a black box that maps $x \mapsto \hat{f}(x)$:

$$x \longrightarrow \boxed{\text{Black Box}} \longrightarrow \hat{y} = \hat{f}(x).$$

DCP instead assumes access to a richer black box that produces an estimate $\hat{F}(\cdot \mid x)$ of the conditional CDF of Y given $X = x$:

$$x \longrightarrow \boxed{} \longrightarrow \hat{F}(y \mid x) \approx \Pr(Y \le y \mid X = x).$$

Notes

(1) Instead of a CDF, one may have conditional *samples* from $Y \mid X$ (e.g., from a generative model). One can still proceed by using these samples to form an empirical $\hat{F}(\cdot \mid x)$.

(2) $\hat{F}(\cdot \mid x)$ may be *inaccurate* — a key motivation for conformalizing it, so that marginal coverage holds regardless of how well $\hat{F}$ approximates the truth.

5.2 The DCP Procedure

Conformalize the CDF estimate by using $\hat{F}$ to define a score function, then plug into split CP. A canonical choice is

$$S(x, y) = \left| \hat{F}(y \mid x) - \tfrac{1}{2} \right|.$$

The motivation for this score is the probability integral transform: if $\hat{F}$ equals the true conditional CDF F, then $\hat{F}(Y \mid X) \sim \text{Uniform}[0, 1]$ independently of X, so $S(X, Y)$ has the same distribution at every X. A single calibration threshold $\hat{q}_{1-\alpha}$ then gives approximately correct coverage at every test point. Unpacking the level set $S(x, y) \le \hat{q}_{1-\alpha}$,

$$C(X_{n+1}) = \left\{ y \in \mathcal{Y} \,:\, \tfrac{1}{2} - \hat{q}_{1-\alpha} \le \hat{F}(y \mid X_{n+1}) \le \tfrac{1}{2} + \hat{q}_{1-\alpha} \right\},$$

i.e., the prediction set consists of the values y whose $\hat{F}$-quantile lies within $\hat{q}_{1-\alpha}$ of the median.

5.3 Guarantees

Suppose $\hat{F}$ is "good" in the following averaged sense:

$$(*) \qquad \Pr_{X_{n+1}}\Big[\big\| \hat{F}(\cdot \mid X_{n+1}) - F(\cdot \mid X_{n+1}) \big\| \le \delta \Big] \ge 1 - \delta.$$

Then DCP enjoys two guarantees:

(1) Approximate conditional coverage. Under $(*)$,

$$\Pr\big[Y_{n+1} \in C(X_{n+1}) \,\big|\, X_{n+1} \big] \ge 1 - \alpha - o(\delta).$$

(2) Marginal coverage always. Regardless of the quality of $\hat{F}$,

$$\Pr_{(X_i,Y_i)_{i=1}^{n+1}}\big[\, Y_{n+1} \in C(X_{n+1}) \,\big] \;\geq\; 1-\alpha,$$

The point is that marginal coverage is "free" — it comes from exchangeability alone via Theorem 3 — while conditional coverage degrades gracefully as $\hat{F}$ departs from the truth. This is the best we can hope for given the impossibility result above.

6 Efficiency

Coverage alone is not enough: the trivial predictor $C(X_{n+1}) = \mathcal{Y}$ achieves coverage 1 but is useless. We also want the prediction set to be *small*.

Definition 9 (Efficiency)**.** Ideally we want $C(X_{n+1})$ to be small in *volume* (e.g. cardinality for discrete $\mathcal{Y}$).

Two views of efficiency. The size of $C(X_{n+1})$ can be interpreted in two complementary ways:

- **Efficiency of the method.** A good underlying black-box model $\hat{f}$ (or $\hat{F}$) yields small prediction sets on average.
- **Measure of unreliability.** The size of $C(X_{n+1})$ at a particular X_{n+1} reflects, in a calibrated way, the model's uncertainty at that point. Large sets flag inputs on which the black box is unreliable.

References

Victor Chernozhukov, Kaspar Wüthrich, and Yinchu Zhu. Distributional conformal prediction. *Proceedings of the National Academy of Sciences*, 118(48):e2107794118, 2021. doi: 10.1073/pnas.2107794118. URL https://arxiv.org/abs/1909.07889.

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